

School Management System

Team Members

Name	Student ID
Marwan Khaled	23-101296
Youssef Idrees	23-101077
Hany Yasser	23-101107
Omar Elsayed	23-101091
Yassin Sameh	23-101192

1. System Overview and Main Goals

Our project is a School Management System that helps schools handle their daily work in an easier and faster way. It allows admins, teachers, and students to use one system instead of many different tools or papers.

Main Goals:

- Make it easier to manage student and teacher information.
Provide students with access to study materials online.
- Keep attendance and grades in one place.
- Help teachers and students communicate better.
- Save time by reducing manual work.
- Make data organized and easy to find.

2. Key Features and Target Users

Features:

- Login for admins, teachers, students and parents.
- Add and update student and teacher details.
- Attendance record for the classes
- Teachers can enter grades; students and parents can see them.
- Admins and teachers can make announcements to users.
- Parents track their children progress according to their grades.
- Generate simple reports for attendance and grades.
- Uploading study materials so students can download or view materials anytime.

Target Users:

- Admin: Controls the whole system and manages accounts.
- Teachers: Add grades, attendance, and class info.

- Students: View grades, schedule, and school news.
- Parents: view attendance, grades, and school announcements related to their children

3. Problem and Value

Problem: Many schools still use paper or old systems to record student data, attendance, and grades. This takes time, causes mistakes, and makes communication slow.

Value: Our system helps schools save time, stay organized, and make communication easier. Everything will be stored digitally and can be accessed anytime. Students can easily access all learning materials in one place, reducing confusion and saving time.

4. Scope and Limitations Scope:

- Manage students, teachers, and admin information.
- attendance, grades, and announcement modules.
- Allow different user roles (admin, teacher, student).
- Accessible through a web browser.
- class scheduling
- communication module between teacher and students • A module for teachers to upload study materials for students.

Limitations:

- It will not include school fees or payment systems.
- There will be no mobile app for now.
- Internet is required to use it.

5. GitHub Repo Link

<https://github.com/Yousuf-Idrees/School-Managment-System.git>

Stakeholders Categories	Identified Stakeholders	aspects of the system particular interest
Internal Stakeholders	• School administrator • Admins • Teachers	• Data management for students and teachers. • Classes schedules and announcements for students. • Attendance and grades. • Uploading study materials.
External Stakeholders	• Parents • Software developers	• Parents: having access to their children's grades and attendance. • Software Developers: Upgrading the system features and keeping the website secured.
Operational Stakeholders	• Students • Teachers • Parents	• Students: check their grades, attendance, announcement and access to the learning resources. • Teachers: entering grades, attendance and sharing the course materials. • Parents: Keeping track of their children's progress and attendance
Executive Stakeholders	• School principal • Vice principal • Head of departments	• Tracking teachers' efforts toward students. • Making decisions according to system data. • Reviewing students' academic levels, grades and attendance to see the school progress.

Functional Requirements

1. User Authentication and Access Control

- 1:** The system shall allow users (Admin, Teacher, Student) to log in using a unique username and password.
 - 2:** The system shall verify user credentials before granting access.
 - 3:** The system should provide different access rights based on user roles:
 - Admin: Full access to manage all modules.
 - Teacher: Access to classes, attendance, and grades.
 - Student: Access to personal grades, schedule, and announcements.
 - 4:** The system shall allow the admin to add, update, or remove user accounts.
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2. Student Management

- 1:** The system shall allow the admin to add, view, edit, and delete student information (e.g., name, ID, class, contact details).
 - 2:** The system allows teachers to view student lists for their classes.
 - 3:** The system shall store all student data securely in a central database.
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3. Teacher Management

- 1:** The system shall allow the admin to add, view, edit, and delete teacher profiles.
 - 2:** The system shall assign teachers to specific classes or subjects.
 - 3:** The system allows teachers to update their profile information (except roles or permissions).
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4. Attendance Management

- 1:** The system allows teachers to record attendance for each class session.
 - 2:** The system shall allow the admin to view attendance records for all classes.
 - 3:** The system shall allow students to view their own attendance history.
 - 4:** The system shall generate attendance reports based on selected dates or classes.
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5. Grades Management

- 1:** The system shall allow teachers to enter and update student grades for each subject.
 - 2:** The system shall calculate and display overall grades or averages automatically (if applicable).
 - 3:** The system shall allow students to view their grades securely.
 - 4:** The system shall allow admins to generate grade reports for review.
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6. Announcements and Communication

- 1:** The system shall allow the admin to create and send announcements to all users or selected groups (e.g., teachers, students).
 - 2:** The system shall display announcements on the dashboard of each user.
 - 3:** The system shall allow teachers to post class-related notices viewable by their students.
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7. Reports and Data Management

- 1:** The system shall generate simple reports on attendance, grades, and student performance.
 - 2:** The system should provide search and filter options to quickly find records.
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8. System Management and Security

- 1:** The system shall automatically back up data periodically.
 - 2:** The system shall log on to user activities (e.g. login, data updates).
 - 3:** The system shall restrict unauthorized access to sensitive data.
 - 4:** The system should provide password reset functionality for users.
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9. System Accessibility

- 1:** The system should be accessible through a standard web browser.
- 2:** The system shall support multiple devices (desktop, laptop, tablet).
- 3:** The system should require an active internet connection to function.

Requirements Elicitation Techniques

Purpose

The goal of this section is to explain how our team will collect the needed information to understand what users want from the School Management System.

Our project includes modules for Class Scheduling and Communication between Teachers and Students, along with other core functions like attendance, grades, and announcements.

We chose a mix of five techniques to make sure we get complete and correct requirements from all users the admin, teachers, and students.

Selected Techniques and it's Justifications:

1. Semi-Structured Interviews

- **Description:**

We will conduct short interviews with the school principal and some teachers and admin staff.

- **Justification:**

This technique helps us collect detailed information about how the school works now and what problems they face.

Through interviews, we can ask about scheduling rules, data management, access control, and communication needs.

It also lets us ask follow-up questions for more details or examples.

The interview agendas include clear questions about user roles, current issues, and improvement ideas.

2. Questionnaires

- **Description:**

We will give questionnaires to teachers and students to collect opinions from many users quickly.

- **Justification:**

The questionnaire covers system access, interface preferences, attendance input, grade viewing, announcements, and communication needs.

This helps us understand how users want to interact with the system and which features are most important to them.

It also supports our interview results by adding real user data and statistics.

3. Document Analysis

- **Description:**

We will review existing documents like class schedules, attendance sheets, grade records, and teacher lists.

- **Justification:**

These documents show how the school currently manages students, teachers, attendance, and grading.

By studying them, we can find what data the system must store, how it is organized, and what problems exist in the manual process.

It also helps us set the rules and structure for the database and scheduling logic.

4. Observation

- **Description:**

We will watch how teachers and admin staff perform their daily tasks such as taking attendance, entering grades, and checking schedules.

- **Justification:**

Observation helps us see the real workflow instead of just hearing about it.

It shows where users face difficulties, which steps take the most time, and which actions can be automated in the system.

It also helps confirm whether the information from interviews is accurate in practice.

5. Prototyping

- **Description:**

We will create simple screen mockups or wireframes for main pages like the dashboard, attendance form, and communication panel.

- **Justification:**

Prototyping allows users to see what the system will look like and give feedback early.

This helps us improve the design before real development begins.

It is especially useful for the class scheduling and teacher-student communication modules, where the interface is important for ease of use.

Interview Agenda

1. Internal Stakeholders (Users) Interview Questions :

A. Teachers (Focus: Attendance, Grades, Communication)

- **Attendance Workflow** : Describe your current step-by-step process for taking attendance for one class. What information do you record, and where is it submitted or stored?
- **Grade Entry Pain Points** : When entering or updating student grades, what currently takes the most time or causes the most frustration? What specific calculations (e.g., weighted averages) do you need the system to handle automatically?
- **Student/Teacher Information Access** : What student details (their name/ID/grades/attendance) do you frequently need to access, and how quickly do you need this information?
- **Communication Needs** : How do you currently send messages or updates to your students? What features would help you "communicate better" within the system?
- **Data Organization** : How important is it that your data (grades and attendance records) is organized and easy to find?

B. Students (Focus: Grade Viewing, Schedule, Announcements)

- **Grade Viewing Experience** : When you want to check your grades, what specific information are you looking for (e.g., individual assignment scores, final course average, missing work)?
- **Announcement Consumption** : How do you currently receive important school news or announcements? What's the best way for the system to

deliver announcements to you (in-system notification (alert on the website) or an email notification)?

- **System Access** : When you access your information (like grades or schedule), how fast must the page load before you become frustrated?

- **Missing Features** : If you could add one capability to the new system to help you manage your school life better, what would it be?

2. External Stakeholders (Client/School) Interview

Questions :

A. School Administrator / Principal / Owner (Focus: Management, Reporting)

- **Data Management Goals (Value/Goals)** : Beyond just "easier management," what are the top 1-2 strategic benefits your school expects to gain from this system (e.g., better parent engagement, audit compliance, increased efficiency)?

- **Report Customization** : For attendance and grades, what specific reports are essential for administration? Do you need to see trends, comparative data (e.g., class vs. class), or just simple individual summaries?

- **User Onboarding/Offboarding (Security)** : Describe the process for creating, updating, and disabling user accounts (Admin, Teacher, Student). Who approves these changes, and how quickly must the system reflect them?

- **Data Migration and Organization** : You mentioned currently using "paper or old systems". Approximately how many years of historical student and teacher data will need to be imported into the new system, and how critical is it that this transition is error-free?

- **Scope & Future Needs** : If the project is successful, would you like to make this software as a mobile application or as a website?

Questionnaire

Internal Stakeholder Questionnaire (Teachers & Students)

A. General and Access (All Users):

- 1- How often during a school day would you need to access the School Management System?
 - Hourly
 - Every few hours
 - Once a day
 - A few times a week

- 2- If the system is web-based, where would you primarily access it from?
 - Desktop/Laptop (at school or home)
 - Tablet/iPad (e.g., using a tablet to take attendance in class)
 - Mobile Phone Browser (accessing the website on a smaller screen)

- 3- When you log in, which dashboard view would be most helpful to see immediately?
 - List of upcoming announcements and assignments
 - Quick links to attendance/grade entry pages
 - A summary of your personal account status

- 4- To ensure the system is "easy to use", which is the single highest priority?
 - **Speed:** Making sure tasks require the fewest steps possible
 - **Looks:** Having a clear, modern, and visually appealing layout
 - **Guidance:** Providing simple labels and help text so you know exactly what to do

B. Teacher-Specific: Attendance, Grades, and Data Management:

5- When you record attendance, which input method would be most efficient?

- List-based check box for all students (fastest for full attendance)
- Quick search for absent students (fastest for low absences)
- Integration with a seating chart diagram for visual input

6- How would you prefer to see the scores weighted for the final average calculation?

- Pre-defined categories with fixed weights (e.g., Homework 20%, Tests 50%)
- Customizable weights that the teacher can set for each individual assignment
- Total points system

7- How often do you need to generate a quick, on-demand grade report for a single student?

- Once per grading period
- Monthly or bi-weekly
- Weekly or more often

8- Which format would you most prefer for downloading and exporting your final grade and attendance data for offline record keeping or transfer to other systems?

- **CSV** : A simple text file, easy to import into any spreadsheet or database program
- **Excel** : A formatted file that retains column headers, specific formulas, and presentation styles.
- **PDF Document** : A secure, read-only document best used for final archival and printing records

C. Student-Specific: Information Viewing and Communication:

- 9- When checking class schedules and assignment due dates, which format would be most usable for you?
- A simple daily/weekly list view
 - An interactive calendar view
 - A graphical timeline showing progress on assignments
- 10- How should the system organize grades for the student view?
- **By Course:** Grouped by current subject
 - **By Due Date:** Ordered chronologically by when the assignment was due
 - **By Category:** Grouped by assignment type
- 11- What specific type of schedule information is most important to view in the system?
- Current Day's Schedule
 - Teacher Contact Information for each class
 - Links to Class Materials or Homework
- 12- How quickly do you expect your latest grade to appear in the system after a teacher says they have finished marking it?
- Immediately
 - Within the same day
 - Within 24 hours

D. Communication and Announcements:

- 13- When receiving urgent school announcements, do you prefer an in-system notification or an email notification?
- Instant Email notification
 - In-system banner/pop-up alert upon login
 - SMS/Text message to a verified mobile number

14- For non-urgent communication (e.g., reminders about homework), which method would you prefer the system to use?

- Standard Email
- Notification feed on the dashboard
- In-system message board

15- If you need to search for a past announcement or a student/teacher record, which search functionality would be most valuable?

- Basic keyword search (finds the exact word or phrase)
- Advanced filter search (by date, subject, user role)
- Contextual search (suggests relevant results as you type)

16- What is the most important feature the system should have to help students and teachers communicate better?

- Built-in direct messaging functionality
- A shared announcement board for each class
- Automated email reminders for deadlines

17- When receiving an announcement about a school event, what is the most essential piece of information you need to see first?

- The Date and Time of the event
- The Date and Time of the event
- The Name and Purpose of the event

18- If an assignment is missing or past due, how should the system inform you?

- A simple red flag/icon next to the assignment name
- A daily email summary of all missing work
- A constant pop-up alert until the work is submitted

E. System Operations and Efficiency:

19- If the internet connection is temporarily lost while entering data, which system behavior is most critical?

- The system saves the data locally and synchronizes once reconnected
- The system issues a clear warning and prevents data entry until reconnected
- The system simply notifies the user and allows them to continue filling out the form

20- For organizing teacher information (schedules, classes, contact info), which view is most important for a teacher's workflow?

- A list of all classes they teach for the current semester
- A monthly calendar showing all their class times and meetings
- A printable, summary schedule page



